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MILITARY SPECIFICATION

CHEMICAL AGENT, PLASTICIZED WHITE PHOSPHORUS (PWP)

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-C-337B. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of plasticized white phosphorus (PWP).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.
TT-X-916 - Xylene (For Use in Organic Coatings).

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).
MIL-R-51209 - Rubbers, Synthetic, For Chemical Agent Compounding.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 1365

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-179 - Department of Transporation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Department of Transportation regulations are a part of the Code of Federal Regulations available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 171-179.")

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D95-62 - Water in Petroleum Products and Other Bituminous Materials.
D1193-66 - Reagent Water.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Composition. PWP shall be compounded from a granulated white phosphorus water slurry and plasticizing mixture in such a manner that the PWP shall be a homogeneous mixture consisting of 75 ± 3 parts by weight of granulated white phosphorus and 25 ± 3 parts by weight of plasticizing mixture.

3.1.1 Granulated white phosphorus. The granulated white phosphorus used to compound PWP shall conform to MIL-C-215 and when in the water slurry before mixing with the plasticizing mixture, shall be in such a state of subdivision so that no less than 98 percent by weight shall pass through a 4760-micron (No. 4) sieve and no less than 40 percent by weight shall pass through a 590-micron (No. 30) sieve when tested as specified in 4.2.4.1.

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3.1.2 Plasticizing mixture. The plasticizing mixture used to compound PWP shall consist of 35 + 1 parts by weight of synthetic rubber conforming to type I or type II of MIL-R-51209 and 65 + 1 parts by weight of xylene conforming to grade B of TT-X-916 (see 6.2).

3.2 Water content. The water content of the PWP shall be no greater than 16 percent by weight when tested as specified in 4.2.4.2.

3.3 Thermal stability. The separation of white phosphorus from the PWP shall be no greater than 5 percent by weight when tested as specified in 4.2.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the PWP offered for acceptance at one time, which has been produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of preparation for delivery. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For white phosphorus water slurry test. A representative specimen of approximately 1/2 pint of white phosphorus water slurry shall be taken from each batch or lot of white phosphorus at the time of transfer to the compounding mixer and placed in a clean, dry, sealed jar labeled to identify the batch or lot from which it was taken (see 6.5).

4.2.2.3 For PWP tests. A representative specimen of approximately 1 pint of PWP shall be taken from each batch or lot of PWP at the filling extruder and placed in a 1-pint clean, dry jar. The remaining air in the jar shall be displaced with water. The jar shall then be sealed and labeled to identify the lot or batch of PWP from which it was taken.

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4.2.3 Inspection procedure.

4.2.3.1 For examination of preparation for delivery. The sample unit shall be one filled shipping container, ready for shipment. Sample containers and the preparation for delivery thereof shall be examined in accordance with MIL-STD-105 and the following classification of defects:

Categories		Defects
Critical:		None defined
Major:		AQL 1.5 percent defective
	101	Contents per container not as specified
	102	Container not as specified
	103	Container closure not as specified
	104	Container damaged or leaking
	105	Quantity of nitrogen or water incorrect
	106	Marking incorrect, missing, or illegible

4.2.3.2 For white phosphorus water slurry test. The specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4.1. If the specimen fails the test, the lot or batch of white phosphorus water slurry represented shall not be used to compound PWP.

4.2.3.3 For PWP tests. The specimen taken in 4.2.2.3 shall be tested as specified in 4.2.4.2 and 4.2.4.3. Failure of either test shall be cause for rejection of the lot or batch of PWP represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Granulation characteristics of white phosphorus water slurry.

(a) Granules weight determination. Fill a graduated cylinder to the top graduation with tap water and weigh to the nearest 0.5 gram (g). Empty the graduate and fill to the top graduation with the white phosphorus water slurry. Weigh to the nearest 0.5 g. Calculate the weight of white phosphorus in the graduate as follows:

$$\text{Weight of white phosphorus in grams} = 1.83\ (B-A)$$

where: A = Weight of graduate filled with tap water in grams, and
B = Weight of graduate filled with white phosphorus water slurry in grams.

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(b) Procedure. Transfer the entire contents of the specimen jar into a 500-milliliter (ml) graduated cylinder and determine the weight of the white phosphorus as specified in (a). Using sieves conforming to RR-S-366, nest a No. 4 sieve on top of a No. 30 sieve. Place a loosely fitting pan on the bottom. Place the nest of sieves under a stream of tap water. Transfer the entire specimen from the 500-ml graduate to the top sieve. Move the sieves under the stream of tap water until as many granules of white phosphorus as possible have been washed through each sieve. Keep the granules wet at all times. Transfer the residue remaining on each sieve to a separate tared 250-ml graduated cylinder by means of a wash bottle and powder funnel. Determine the weight of residue on each sieve as specified in (a). Calculate the percent by weight passing through each sieve as follows:

$$\text{Percent passing through No. 4 sieve} = \frac{100\ (W - A)}{W}$$

$$\text{Percent passing through No. 30 sieve} = \frac{100\ (W - A - B)}{W}$$

where: A = Weight of residue retained on No. 4 sieve in grams,
B = Weight of residue retained on No. 30 sieve in grams, and
W = Weight of white phosphorus in specimen in grams.

4.2.4.2 Water content. Determine percent water content in accordance with ASTM D95 using approximately 25 g of specimen and xylene conforming to grade B of TT-X-916 as the solvent.

4.2.4.3 Thermal stability. Weigh a 60-ml large mouth, screw top bottle and its cover to the nearest 0.1 g. Transfer approximately 20 to 30 g of the specimen to the bottle, displace the remaining air with carbon dioxide, cover, and weigh to the nearest 0.1 g. Place in a constant temperature bath at 67° to 73° C for 2 hours and then allow to cool to 20° to 30° C. If any of the white phosphorus has separated from the original mass of PWP, remove the separated white phosphorus and determine by difference the weights of the separated white phosphorus and the PWP. (If desired, the PWP may be placed in a small glass cup approximately 1 inch high and 1-1/2 inches in diameter which has slotted openings around the circumference of its convex bottom and is supported by three small glass legs. The cup is then placed inside the 60-ml bottle. Any separated white phosphorus will drip through the cup slots after heating and cooling as above. The bottle may then be submerged under water and its contents covered by water. Separated white phosphorus may be removed with a pair of forceps and quickly blotted to remove excess water and then weighed in a tared beaker containing water.) Calculate the percent white phosphorus separated as follows:

$$\text{Percent white phosphorus separated} = \frac{100\ A}{W}$$

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where: A = Weight of white phosphorus separated in grams, and
W = Weight of specimen in grams.

5. PREPARATION FOR DELIVERY

5.1 Packing, level C. Unless otherwise specified in the contract or order (see 6.2), PWP shall be packed in a 30-gallon capacity steel drum conforming to Department of Transportation (DOT) Specification 6D, furnished with completely welded bottom chime and "U" rolling hoops. Immediately after filling the drum with PWP, either the surface of PWP shall be completely blanketed with a layer of water no less than 1-inch thick, or all residual air in the drum shall be displaced with nitrogen, and then the drum shall be immediately closed. A void of no less than 3-percent by volume, including water if used, shall be allowed in the filled and closed drum.

5.2 Marking. In addition to the marking specified in 5.2.1 and any special marking required by the contract or order (see 6.2), drums shall be marked in accordance with DOT regulations and MIL-STD-129.

5.2.1 Precautionary marking. Each drum shall bear the DOT yellow "Flammable Solids and Oxidizing Materials" label and shall be durably and legibly marked as follows:

PLASTICIZED WHITE PHOSPHOSPHORUS
DANGER! CAUSES SEVERE BURNS
POISON

Contents are packed under nitrogen or water and will ignite if nitrogen or water is removed.
Do not get on skin or in eyes.
Wear heavy rubber gloves and goggles or face shield.
In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes. Keep skin area wet until medical attention is obtained.
Keep container out of sun and away from heat.
Empty containers must be filled to their entire capacity with water or nitrogen and the DOT "Caution-Residual Phosphorus" placard affixed thereto.

6. NOTES

6.1 Intended use. PWP is intended for use in filling white smoke munitions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Packing other than that specified, if required (see 5.1).

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(c) Additional marking, if required (see 5.2).
(d) Whether type I or type II rubber is required (see 3.1.2).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Sampling and testing of white phosphorus. Containers of white phosphorus should contain at least 3 inches of water above the surface of the white phosphorus. White phosphorus is a highly flammable substance with an exceedingly low autoignition temperature. Every precaution should be taken during manufacture, storage, sampling, and testing to minimize the danger to personnel and equipment.

Custodian:

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Preparing activity:

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Project No. 1365-A037

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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